# AES (Advanced Enviro-Septic$^{TM}$) Owners Manual

**New Zealand Distributors**
Environment Technology Ltd
105 Pascoe Street
Nelson 7010
(03) 9707 979
www.et.nz
info@et.nz

"Always The First Option"
ADVANCED ENVIRO-SEPTIC™

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## Technical support

Environment Technology provides technical support to all individuals and companies using AES and other Presby Environmental products. For questions about products or information in this manual please contact us at 03 9707 979, info@et.nz

## Important Safety Information

- Please ensure that the cover/s of the septic tank, the pumping station and sampling device if installed, are always in place and that they remain accessible at all times for periodic inspections and interventions when necessary.
- Ensure you receive an accurate As-Built plan of your system from your installer. Pipes are buried near your septic installation. Please speak to your installer or consult the as-built plan prior to digging or excavating near your septic system.
- It can be dangerous even potentially deadly to open a septic tank, pumping station or any enclosed space that is part of a wastewater treatment system. The action of the bacteria on the organic matter present in the wastewater produces gases such as carbon dioxide ($CO_2$), methane ($CH_4$) and hydrogen sulphide ($H_2S$). The $H_2S$ present in the septic tank or a pumping station can cause the death of an individual in a matter of minutes. A well-maintained ventilation system will reduce the risk of toxic gases build up, however work in this area must be carried out by competent personnel.

# Introduction

Thank you for choosing the AES system for your septic installation. This system was developed to efficiently treat domestic wastewater. Instructions must be followed in order to maintain its treatment performance so that you can make use of it for many years. Carefully read through this entire document and retain it in your files for future reference.

**The purpose of this document**

This user guide explains the proper use, procedures and inspections required in order to ensure the proper operation of your AES system for residential wastewater treatment.

It is the owner's responsibility to ensure that the system is used properly and according to its treatment capacity. It is also their responsibility to respect the rules and regulations in effect regarding associated council and government regulations.

**Designation of the AES System**

**Name**: AES Wastewater System

**Application Domain**: Residential Wastewater (sewage).

**Class and treatment type**: The AES system meets all the performance criteria requirements of both the Australian/ New Zealand Standard AS/NZS 1546.3: 2008, and the Queensland Plumbing and Wastewater Code: 2011 (for both Secondary and Advanced Secondary treatment). In 2017 AES completed Trial 12 of the Onsite Effluent Treatment System (OSET) National Testing Facility in Rotorua which certified secondary treatment quality.

The system cannot be used to treat wastewater to make it consumable. It is made to treat residential wastewater, and some commercial wastewater to an acceptable level for it to be reintroduced into the environment.

**Definition of the AES System**

The AES system is composed primarily of two inseparable components: the rows of AES pipe and a layer of system sand.

The AES system must be preceded by a septic tank or equivalent primary treatment system. The treated water is generally drained directly into the soil beneath the treatment system through a soil absorption system.

**What to do if a problem occurs?**

If in the course of normal use of your septic system you notice any of the following problems:

- Abnormally wet soil, presence of persistent puddles or odours in the area of the septic tank or the AES system,
- Slow flushing toilets or other plumbing in the home,
- Presence of abnormally abundant vegetation on the surface or around the septic tank or the AES system installation,
- Flooding in the area where the AES system is installed,
- Erosion of the land fill on or around the AES system,
- Alarm from the pumping station if such a device is part of your installation.

Please contact your AES certified contractor or Environment Technology. There are often simple remedies.

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| **Customer Service and Technical Support information** | Please do not hesitate to contact us if you need further information.<br>**Environment Technology can be contacted at:**<br>**Telephone: 03 970 7979**<br>**Email: info@et.nz**<br>**Website: www.et.nz**<br>**Address: 14 Onekaka Iron Works Rd, Takaka 7182** |
| **Certified Contractor** | The AES System must be installed by a licensed drainlayer with AES certification. Certification is obtained by successfully completing the online AES Certification Course. This course can be accessed at www.et.nz<br><br>Environment Technology can provide the name of drainlayers having the proper certification to install AES systems. This information is also available on our website http://www.et.nz/installers/ |
| **AES System Capacity** | The capacity of the AES System depends on two elements:<br>• The number of AES Pipes<br>• The capacity of the underlying soil to evacuate the treated water<br><br>The total volume of wastewater fed to the system must not be more than what is shown in the design. The design flow is generally a weekly average.<br><br>The system may also be limited by the capacity of the underlying soil to permit the infiltration and evacuation of wastewater. This value is evaluated by the designer who created the plans and estimate for your septic installation. The design should take into account whether the capacity of the soil is a potential limiting factor. |
| **Warranty certificate** | AES comes with a 20 year manufacturer's limited warranty. The warranty details are presented in Appendix A (page 16) |

## Functioning of the AES System

The AES system is a passive technology which facilitates the proliferation of the aerobic bacteria responsible for wastewater treatment. It is comprised mainly of two inseparable components: the rows of AES pipes and a layer of system sand.

The AES system must be preceded by a septic tank or equivalent primary treatment system.

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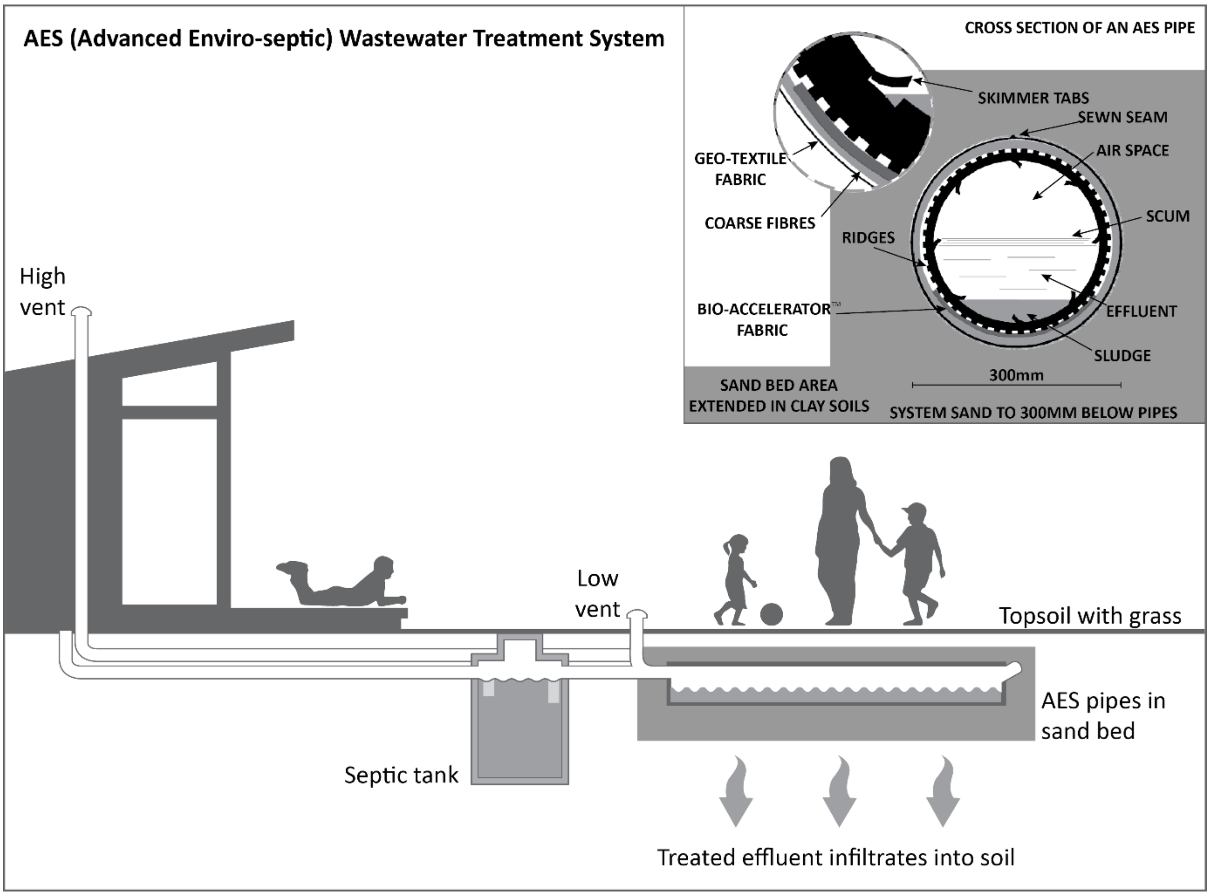
| **Treatment process of AES** | The rows of AES pipes and system sand permit the treatment and distribution of wastewater on the surface of the receiving soil.<br><br>The pipes support, first of all, the separation of particles through flotation and decantation. The water is then evacuated through perforations situated around the pipes and through the pores of the two layers of synthetic media covering the pipes. These membranes facilitate the fixation of the microbial cultures which support wastewater treatment, as well as longitudinal distribution of the effluent.<br><br>The layer of sand continues the treatment process and helps to disperse the water before it infiltrates into the natural soil. |

# Diagram of the AES system




# AES system components

Your septic installation includes several components.  All of these components are parts of the chain of

treatment of your installation. The following table presents a list of these elements.  However, it should be noted that some of these are only used when site conditions require them.  The table also presents a summary of inspections required for each component.  More detailed information on this subject is presented in the sections that follow.

## Table: AES System Components

| **Component of the septic system** | **Function** | **Follow-up needed** | **Frequency** | **Responsible for follow-up** |
|---|---|---|---|---|
| Septic tank | Primary wastewater treatment | Periodic emptying | According to standards and regulations in effect | Owner is respon-sible to have work done by qualified person |
| Septic Tank Effluent Filter* | Retention of solids in low pressure pumped applications. | According to manufacturer's instructions. | | |
| Distribution systems if required for larger dual bed systems.<br>3 options<br>A) Gravity Distribution box and flow equalizers | Distributes the septic tank effluent to the rows of AES | A) According to the water level in the inspection port | A) As needed | A) Owner |
| B) Pressure distribution (pump) system | | B) According to the manufacturer's directions. | | |
| C) Automatic distributing valve | | C) According to the manufacturer's directions. | | |
| Rows of AES pipes | Treat and distribute effluent | | | |
| Sampling device | To verify the treatment performance of the AES System | Ensure that there is access to this device | Optional | Qualified person |
| Vent | To allow the circulation of air in the AES System | Ensure that the opening is not blocked | As needed | Owner |
| System sand | To complete the water treatment process and to improve the drainage | No | | |
| Pumping station (optional) | Lift septic tank effluent to the AES System | According to supplier's specifications | | |

*The effluent filter is necessary whenever the septic tank is followed by a pump distribution system.

## Operating the AES System

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| **Initial Use** | At the time of installation the septic tank should be filled with clear water.<br>If a pumping station is used, the contractor will verify that it is functioning properly at the time of installation. The home owner must make sure that there is adequate electricity to safely operate the equipment as well as the alarm component.<br>The AES system is now ready for use. |
| **Intermittent Use or Prolonged Absences** | The AES system is a passive wastewater treatment system. When properly installed, it requires no particular attention even if you are away for periods of time. |

## AES System Operating Instructions

The use and maintenance of AES Systems are relatively simple. In general, respecting the following rules will allow you use of your system without problems for years to come.

**Wastewater Volume**

Excessive quantities of water that leave the house and enter the AES System in a short period of time could have a negative impact on the effectiveness of the treatment and the infiltration of wastewater causing agitation in the septic tank. A quantity of sludge or scum is likely to be put into suspension and be brought towards the system and the infiltration bed.

After the installation, if changes are made to the residence (eg. addition of a bedroom), please contact the designer of the AES System. Make sure that the septic system is inspected by a qualified person to determine that it has the necessary capacity to treat and infiltrate the new daily design flow of wastewater being generated.

**In the bathroom**

**Do:**

- Immediately repair any leaking tap or toilet,
- Use a reasonable quantity of toilet paper.
- Minimise or avoid bleach, antiseptic disinfectants, and ammonia acids in the system

**Do not :**

- Use disinfectant in tablet (puck) form, whether it is placed in the basin or the tank,
- Throw cigarettes, cigarette butts or medication in the toilet,
- Throw paper towels, paper napkins or other personal hygiene products in the toilet.

**In the kitchen**

**Do:**

- Repair any leaking tap,
- Use dish soap or dishwasher soap that is low in phosphate (0-5%),
- Use the necessary quantity of soap to do the work. Take note that the necessary quantity is often less than suggested by the manufacturer.
- Use biodegradable soap, low-phosphorus or phosphorus free detergents.

**Do not :**

- Use a food waste disposal unit in your sink that is connected to your septic installation. If you do have a waste disposal unit, your septic tank may require more frequent pump out to remove sludge build up.
- Dispose of vegetables, meats, fat, oil, coffee beans, citrus products or other products into the septic system.

**For the laundry**

**Do:**

- Use phosphate free detergent, preferably in liquid form. If it is not possible, use biodegradable powder detergent,
- Use the necessary quantity of soap to do the work. Take note that the necessary quantity is often less than that suggested by the manufacturer,
- Minimize the volume of water used for the laundry according to the quantity of clothing to wash,
- If possible spread your loads of laundry throughout the week
- Prevent harsh chemicals entering the system (e.g. paint, nappies)

**Elsewhere in and around the house**

**Do:**

- Divert drainage and rain water away from the surface of the AES system.
- Roof and surface water should be redirected away from absorption trenches.

**Do not :**

- Discharge water softener backwash into your septic system,
- Discharge any water from swimming pool filters, spas or other appliances that discharge chlorinated water into your septic system.
- Let water from sump pumps, gutters and drainage pipes discharge into the septic system,
- Dispose of solvents, paints, antifreeze, engine oil or other chemicals in the septic installation. This includes water used to wash brushes or rollers that were used with latex paint (latex paint contains elements that are harmful to septic system),
- Dispose of animal litter in the septic installation.

**Chemicals for septic installation**

Your AES system does not require any starting chemical, cleaning or other additives. The bacteria that carry out the treatment are naturally present in raw domestic sewage. Any chemicals or additives added to the AES System could possibly kill these bacteria.

**Ventilation**

It is very important to ensure that good ventilation occurs so that the septic system functions correctly. The vent(s) installed at the ends of the septic system encourage this air circulation. It is important to make sure that the opening is not blocked and that air can circulate freely at all times. Air enters through the low vent, circulates through the rows of pipes and exits through the high vent.

The owner must be sure to have a roof vent and to keep it clear at all times. When a pumping station is used, a bypass pipe or an extra vent must be used to ensure proper ventilation of the system.

**Heavy machinery & motorized vehicle traffic**

No vehicles or heavy machinery must be driven over a septic tank. Heavy machinery or motorized vehicle traffic on the soil around the AES bed closes the natural pores of the soil which reduces its permeability and allows for ponding and the accumulation of water.

**Vegetation**

The surface of the AES system must be planted with grass or other vegetation that forms a thick turf. This will encourage surface water runoff from the bed surface. The vegetation must be cut regularly in order to encourage growth without the use of fertilisers. Vegetation cover contributes to the elimination of nitrogen and phosphorus.

It is important ***not*** to plant trees or other plants with invasive roots such as figs, willows, blackwood and many others within 3 metres of the AES system installation footprint.

## AES System Maintenance

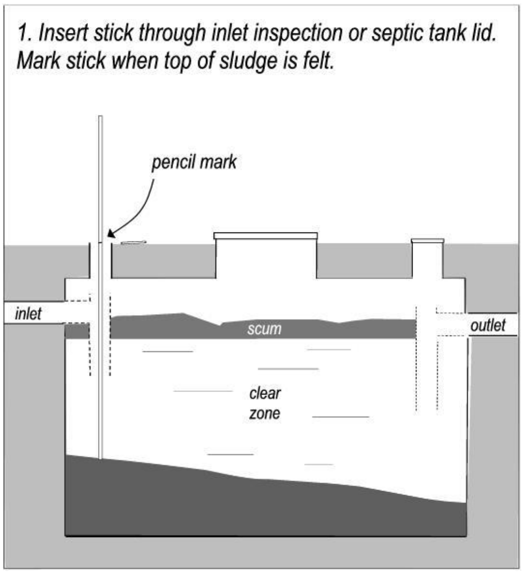
**Septic Tank Maintenance**

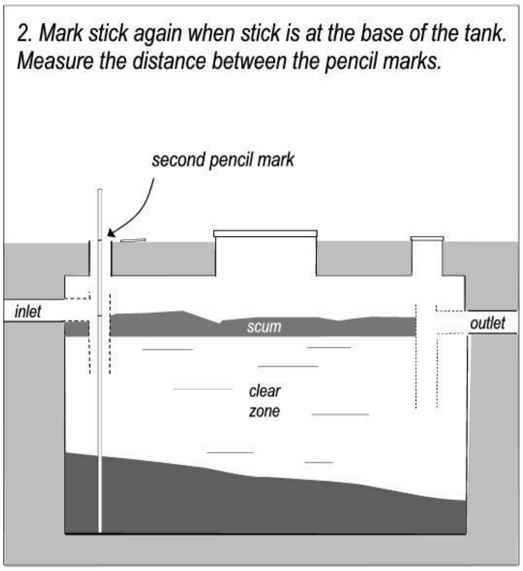
The septic tank preceding the AES System must be pumped out regularly (every 3-5 years for normal residential use or when sludge exceeds 1/3 of the tank volume).

If the septic tank is not emptied regularly, an increasing amount of solids and grease in suspension will leave the septic tank and end up in the treatment system and in time the performance of the AES System may be affected.

The owner must ensure their septic tank is pumped out according to council regulations, if any. This work should always be done by a qualified person.

Note: It is the home owner's responsibility to make sure that at all times the septic tank lids are in their proper position and securely fastened.




**Septic tank outlet (effluent filter)**

An outlet filter is not necessary at the exit of the septic tank in a gravity system. However it must be installed before a pump, for example when pumped effluent is between the septic tank and the AES pipes.

If installed the effluent filter must be cleaned according to the maintenance and inspection procedures provided by the manufacturer.

**AES Pipe Rows**

Under normal use, the rows of AES pipe do not require maintenance. It is normal to find fluctuation of the water level in the pipes. In many installations water level in the pipes can be measured by removing the low vent.

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| **Vent** | The owner must ensure that nothing prevents the circulation of air. There must also be a difference of at least 3 metres, at all times, between the entry vent situated at the extremity of the AES system and the high vent. |
| **System Sand** | There is no maintenance to be done on the system sand during normal use of the AES System. |
| **Pumping station or low pressure distribution system** | In certain cases, the site constraints require the use of a pumping station or a low-pressure distribution system to evenly dispose of the treated effluent. The owner is then responsible to comply with the manufacturer's scheduled maintenance requirements of this equipment. |
| **Embankment surface above the AES System** | The surface located above the AES system must be covered with herbaceous vegetation. A slight slope must be given to the surface in order to help the drainage of rainwater towards the outside of the system. The grass must also be cut regularly. Finally, any depression that could be created with time must be filled in order to avoid any accumulation of water above the system and to prevent erosion. |

## Owner's Responsibilities

**Owner's Responsibilities**

The owner is responsible for:

- Using the AES System according to the instructions presented in this user guide
- Pumping out the septic tank according to the regulations in effect
- Maintaining the effluent filter (if present), the pumping station, the pressure distribution system or the automatic wastewater distributing valve according to manufacturer's specifications and recording the information if this equipment is part of the system
- Ensuring that the vent openings are clear of any obstacle
- Adhering to the requirements of the applicable rules and regulations

**Qualified Person**

Any maintenance of an AES System must be undertaken by a person trained to carry out the inspections of the system, perform adjustments to the equalizers and/or carry out a rejuvenating procedure.

To obtain the name of a qualified person in your area, contact:

**Environment Technology**
**14 Onekaka Iron Works Rd, Takaka 7182**
**info@et.nz**
**03 970 7979**

Information on installers is also available on our website [http://www.et.nz/installers/](http://www.et.nz/installers/)

For maintenance of the pumping station and the low pressure distribution system, if installed, the owner must refer to the user guide specified by the manufacturer of these systems.

The pumping out of the septic tank must be performed by a company specializing in that field.

## Maintenance Sheet

### AES On-site Wastewater Treatment – Passive system

Address:_______________________________________________________Date:______________

Name of AES qualified servicer:______________________________Consent No:______________

**Septic Tank**

Ensure lids of the wastewater treatment system are readily accessible at all times
Measure depth of scum and solids in the septic tank:

Depth of scum: ______

Depth of solids: ______

Depth of tank: ______

Pumping out the septic tank is necessary if solids and scum layers combined are greater than one half the depth of the septic tank.

**AES Bed Venting**

Ensure low vent and high vent are free of vegetation/ restrictions. Yes No

**Notes**

___________________________________________________________________________________

___________________________________________________________________________________

___________________________________________________________________________________

Overall condition of wastewater system, including disposal field:

___________________________________________________________________________________

___________________________________________________________________________________

This report shall be kept by the consent holder. In addition, the consent holder shall also keep written records of all repairs made to any part of the wastewater treatment and land application system.

## Appendix A- Presby Twenty Year Limited Warranty

**PRESBY ENVIRONMENTAL, INC.**
*INNOVATIVE SEPTIC TECHNOLOGIES*

This Twenty Year Limited Manufacturer's Warranty is provided by the Manufacturer, Presby Environmental, Inc., a New Hampshire corporation having a mailing address of 143 Airport Rd., Whitefield, New Hampshire, 03598 (hereinafter called "Presby"). This Warranty applies only to Presby Products sold by or through its duly authorized distributor Chankar Environmental an Australian corporation having a mailing address of Unit 6-62 Rene St, Noosaville, Qld 4566 (hereinafter called the "Distributor"). "Presby Products" means Presby's Advanced Enviro-Septic™ leaching systems and Presby Maze© with the required accessories (couplings, offset adaptor).

**Warranty:** Presby warrants that Presby Products are free from defect for twenty years from the date of installation but in no event more than twenty-one years from the date of manufacture. Product Defects means defects or damage to the Products caused by or occurring during the manufacturing process. This Warranty does not cover or apply to damages to the Products caused by or resulting from transit or from accident, misuse, abuse, neglect, storage, installation, repair, maintenance or from use other than normal and ordinary use of the Products. This Warranty does not apply to damages to the Products caused by or resulting from failure to install or use the Products in accordance with distributor’s instructions which have been approved by Presby or failure to properly inspect and maintain the Products.

**Warranty Registration, Claim Process and Remedy:** Any claim under the Warranty must be in writing and received by the distributor within thirty days of the date when the facts giving rise to such claim under this Warranty become known or are otherwise discovered. The distributor must be provided with an opportunity to inspect the Products as installed. Failure to comply with these requirements renders the Warranty null and void. If, during the Warranty period, the distributor and Presby find and determine that defects in Products exist, then the distributor and Presby's sole and exclusive obligation is to either repair the Products or provide replacement Products. The distributor and Presby, in their discretion, shall determine whether to repair the Products or provide replacement Products. The distributor and Presby shall have no obligation to remove any defective Products or to install any replacement Products. The distributor and Presby shall not be liable or responsible for any other damages or claims arising from or relating to defective Products, including but not limited to claims for general, consequential, or incidental damages, lost profits, or attorney fees.

**Disclaimer:** The distributor and Presby otherwise make no express warranty concerning the Products and the distributor and Presby disclaims any and all warranties, express or implied. Except as stated herein, there are no warranties express or implied, and the distributor and Presby do not warrant that the goods are merchantable or fit for any particular purpose. Any claim or controversy relating to this Warranty, or to matters of place of contracting, interpretation, performance or breach thereof, shall be brought in and adjudged in accordance with the applicable laws of state of New Hampshire.